Name of the product

RESPIDAY Discair 160/4.5 mcg inhalation powder

RESPIDAY Discair 320/9 mcg inhalation powder

Name and strength of Active Ingredient

RESPIDAY Discair 160/4.5 mcg inhalation powder

Each delivered dose (the dose that leaves the mouthpiece) contains: budesonide 160 micrograms/inhalation and formoterol fumarate dihydrate 4.5 micrograms/inhalation.

Each metered dose contains: budesonide 200 micrograms/inhalation and formoterol fumarate dihydrate 6 micrograms/inhalation.

RESPIDAY Discair 320/9 mcg inhalation powder

Each delivered dose (the dose that leaves the mouthpiece) contains: budesonide 320 micrograms/inhalation and formoterol fumarate dihydrate 9 micrograms/inhalation.

Each metered dose contains: budesonide 400 micrograms/inhalation and formoterol fumarate dihydrate 12 micrograms/inhalation.

Dosage form

Oral Inhalation

Product Description

White powder in blister alu foil. RESPIDAY is contained in the inhalation device in Alu / Alu blister packages containing 60 doses of inhalation powder.

Pharmacokinetics:

Absorption

The fixed-dose combination of budesonide and formoterol, and the corresponding monoproducts have been shown to be bioequivalent with regard to systemic exposure of budesonide and formoterol, respectively. In spite of this, a small increase in cortisol suppression was seen after administration of fixed-dose combination compared to the monoproducts. The difference is considered not to have an impact on clinical safety.

There was no evidence of pharmacokinetic interactions between budesonide and formoterol.

Pharmacokinetic parameters for the respective substances were comparable after the administration of budesonide and formoterol as monoproducts or as the fixed-dose combination. For budesonide, AUC was slightly higher, rate of absorption more rapid and maximal plasma concentration higher after administration of the fixed combination. For formoterol, maximal plasma concentration was similar after administration of the fixed combination. Inhaled budesonide is rapidly absorbed and the maximum plasma concentration is reached within 30 minutes after inhalation. In children 6-16 years of age the lung deposition falls in the same range as in adults for the same given dose. The resulting plasma concentrations were not determined.

Inhaled formoterol is rapidly absorbed and the maximum plasma concentration is reached within 10 minutes after inhalation. The systemic bioavailability is about 61% of the delivered dose.

Distribution

Plasma protein binding is approximately 50% for formoterol and 90% for budesonide. Volume of distribution is about 4 L/kg for formoterol and 3 L/kg for budesonide. Formoterol is inactivated via conjugation reactions (active O-demethylated and deformylated metabolites are formed, but they are seen mainly as inactivated conjugates). Budesonide undergoes an extensive degree (approximately 90%) of biotransformation on first passage through the liver to metabolites of low glucocorticosteroid activity. The glucocorticosteroid activity of the major metabolites, 6-beta-hydroxy-budesonide and 16-alfa-hydroxy-prednisolone, is less than 1% of that of budesonide. There are no indications of any metabolic interactions or any displacement reactions between formoterol and budesonide.

Elimination

The major part of a dose of formoterol is transformed by liver metabolism followed by renal elimination. After inhalation, 8% to 13% of the delivered dose of formoterol is excreted unmetabolised in the urine. Formoterol has a high systemic clearance (approximately 1.4 L/min) and the terminal elimination half-life averages 17 hours.

Budesonide is eliminated via metabolism mainly catalysed by the enzyme CYP3A4. The metabolites of budesonide are eliminated in urine as such or in conjugated form. Only negligible amounts of unchanged budesonide have been detected in the urine. Budesonide has a high systemic clearance (approximately 1.2 L/min) and the plasma elimination half-life after i.v. dosing averages 4 hours.

Pharmacokinetic/pharmacodynamic relationship(s)

The pharmacokinetics of budesonide or formoterol in children and patients with renal failure are unknown. The exposure of budesonide and formoterol may be increased in patients with liver disease.

Linearity/non-linearity

Systemic exposure for both budesonide and formoterol correlates in a linear fashion to administered dose.

Pharmacodynamics:

Pharmacotherapeutic group: Drugs for obstructive airway diseases, adrenergics and other drugs for obstructive airway diseases.

ATC code: R03AK07

Mechanism of action and pharmacodynamic effects

RESPIDAY contains formoterol and budesonide, which have different modes of action and show additive effects in terms of reduction of asthma exacerbations. The mechanisms of action of the two substances respectively are discussed below.

Budesonide

Budesonide is a glucocorticosteroid which when inhaled has a dose-dependent anti-inflammatory action in the airways, resulting in reduced symptoms and fewer asthma exacerbations. Inhaled budesonide has less severe adverse reactions than systemic corticosteroids. The exact mechanism responsible for the anti-inflammatory effect of glucocorticosteroids is unknown.

Formoterol

Formoterol is a selective β_2 adrenoceptor agonist that when inhaled results in rapid and long-acting relaxation of bronchial smooth muscle in patients with reversible airways obstruction. The bronchodilating effect is dose-dependant, with an onset of effect within 1-3 minutes. The duration of effect is at least 12 hours after a single dose.

Indications

RESPIDAY Discair 160/4.5 mcg inhalation powder

Asthma

Respiday Discair is indicated for the treatment of asthma, to achieve overall asthma control, including the relief of symptoms and the reduction of the risk of exacerbations (see Section Posology and Method of Administration).

Chronic Obstructive Pulmonary Disease (COPD)

Respiday Discair is indicated in adults, aged 18 years and older, for the symptomatic treatment of patients with COPD with FEV_1 < 70% predicted normal (post bronchodilator) and an exacerbation history despite regular bronchodilator therapy.

RESPIDAY Discair 320/9 mcg inhalation powder

Asthma

Respiday Discair is indicated in the regular treatment of asthma where use of a combination (inhaled corticosteroid and long-acting β_2 adrenoceptor agonist) is appropriate:

- patients not adequately controlled with inhaled corticosteroids and "as needed" inhaled short-acting β2 adrenoceptor agonists

or

- patients already adequately controlled on both inhaled corticosteroids and long-acting β2 adrenoceptor agonists.

Chronic Obstructive Pulmonary Disease (COPD)

Respiday Discair is indicated in adults, aged 18 years and older, for the symptomatic treatment of patients with COPD with FEV_1 < 70% predicted normal (post bronchodilator) and an exacerbation history despite regular bronchodilator therapy.

Recommended dose

RESPIDAY Discair 160/4.5 mcg inhalation powder

Asthma
RESPIDAY Discair 160/4.5 mcg inhalation powder can be used according to different treatment approaches:
A. RESPIDAY anti-inflammatory reliever therapy (patients with mild disease).
B. RESPIDAY anti-inflammatory reliever plus maintenance therapy.
C. RESPIDAY maintenance therapy (fixed dose).

A. RESPIDAY anti-inflammatory reliever therapy (patients with mild disease)
RESPIDAY Discair 160/4.5 mcg inhalation powder is taken as needed for the relief of asthma symptoms when they occur and as a preventative treatment of symptoms in those circumstances recognised by the patient to precipitate an asthma attack. Patients should be advised to always have RESPIDAY Discair 160/4.5 mcg inhalation powder available for relief of symptoms.

Preventative use of RESPIDAY Discair 160/4.5 mcg inhalation powder for allergen- or exercise-induced bronchoconstriction (AIB/EIB) should be discussed between physician and patient; the recommended dose frequency should take into consideration both allergen exposure and exercise patterns.

<u>Adults and adolescents (12 years and older)</u>
Patient should take 1 inhalation of RESPIDAY Discair 160/4.5 mcg inhalation powder as needed in response to symptoms. If symptoms persist after a few minutes, an additional inhalation should be taken. No more than 6 inhalations should be taken on any single occasion.
A total daily dose of more than 8 inhalations is normally not needed, however a total daily dose of up to 12 inhalations can be used temporarily. If the patient experiences a three-day period of deteriorating symptoms after taking additional as needed inhalation, the patient should be reassessed for alternative explanations of persisting symptoms.

<u>Children under 12 years</u>
RESPIDAY anti-inflammatory reliever therapy is not recommended for children.

B. Respiday anti-inflammatory reliever plus maintenance therapy
Patients take a daily maintenance dose of RESPIDAY and in addition take RESPIDAY as needed in response to symptoms. Patients should be advised to always have RESPIDAY available for rescue use.

RESPIDAY anti-inflammatory reliever plus maintenance therapy should especially be considered for patients with:
- inadequate asthma control and in frequent need of reliever medication
- asthma exacerbations in the past requiring medical intervention

Close monitoring for dose-related adverse effects is needed in patients who frequently take high numbers of RESPIDAY as-needed inhalations.

Preventative use of RESPIDAY Discair 160/4.5 mcg inhalation powder for AIB/EIB should be discussed between physician and patient; the recommended dose frequency should take into consideration both allergen exposure and exercise patterns.

Recommended doses:
<u>Adults and adolescents (12 years and older):</u>
Anti-inflammatory Reliever Therapy (in addition to Maintenance Therapy)
Patients should take 1 additional inhalation of RESPIDAY Discair 160/4.5 mcg inhalation powder as needed in response to symptoms to control asthma. If symptoms persist after a few minutes, an additional inhalation should be taken. Not more than 6 inhalations should be taken on any single occasion.

Maintenance Therapy
Patients also take the recommended maintenance dose of RESPIDAY Discair 160/4.5 mcg inhalation powder, which is 2 inhalations per day, given either as one inhalation in the morning and evening or as 2 inhalations in either the morning or evening. For some patients, a maintenance dose of RESPIDAY Discair 160/4.5 mcg inhalation powder 2 inhalations twice

daily may be appropriate. The maintenance dose should be titrated to the lowest dose at which effective control of asthma is maintained.

Maximum Daily RESPIDAY Dose
A total daily dose of more than 8 inhalations is not normally needed; however, a total daily dose of up to 12 inhalations could be used for a limited period. If the patient experiences a three-day period of deteriorating symptoms after taking the appropriate maintenance therapy and additional as needed inhalation, the patient should be reassessed for alternative explanations of persisting symptoms.

Children under 12 years:
RESPIDAY anti-inflammatory reliever plus maintenance therapy is not recommended for children.

C. RESPIDAY maintenance therapy (fixed dose)
When maintenance treatment with a combination of ICS and LABA is required, RESPIDAY Discair 160/4.5 mcg inhalation powder is taken as a fixed daily dose treatment, with a separate short-acting bronchodilator for relief of symptoms. Patients should be advised to have their separate short-acting bronchodilator available for relief use at all times.

Recommended doses:
Adults (18 years and older)
1-2 inhalations twice daily. Some patients may require up to a maximum of 4 inhalations twice daily.

Adolescents (12-17 years)
1-2 inhalations twice daily.

Children under 6 years
As only limited data are available, RESPIDAY is not recommended for children under 6 years old.

Patients should be regularly reassessed by their prescriber/health care provider, so that the dosage of RESPIDAY remains optimal. The dose should be titrated to the lowest dose at which effective control of symptoms is maintained. When control of symptoms is maintained with the lowest recommended dosage, then the next step could include a test of inhaled corticosteroid alone.

In usual practice when control of symptoms is achieved with the twice daily regimen, titration to the lowest effective dose could include RESPIDAY given once daily, when in the opinion of the prescriber, a long-acting bronchodilator in combination with an inhaled corticosteroid would be required to maintain control.

Increasing use of a separate rapid-acting bronchodilator indicates a worsening of the underlying condition and warrants a reassessment of the asthma therapy.

COPD

Recommended doses:
Adults (18 years and older)
2 inhalations twice daily

General information
If patients take RESPIDAY as anti-inflammatory reliever (either alone or in combination with maintenance therapy) physicians should discuss allergen exposure and exercise patterns with the patients and take these into consideration when recommending the dose frequency for asthma treatment.

Special patient groups: There are no special dosing requirements for elderly patients. There are no data available for use of RESPIDAY in patients with hepatic or renal impairment. As budesonide and formoterol are primarily eliminated via hepatic metabolism, an increased exposure can be expected in patients with severe liver diseases.

RESPIDAY Discair 320/9 mcg inhalation powder

Asthma
RESPIDAY is not intended for the initial management of asthma. The dosage of the components of RESPIDAY is individual and should be adjusted to the severity of the disease. This should be considered not only when treatment with combination products is initiated but also when the maintenance dose is adjusted. If an individual patient should require a combination of doses other than those available in the combination inhaler, appropriate doses of β2 adrenoceptor agonists and/or corticosteroids by individual inhalers should be prescribed.

Recommended doses:
Adults (18 years and older)
1 inhalation twice daily. Some patients may require up to a maximum of 2 inhalations twice daily.

Adolescents (12-17 years)
1 inhalation twice daily.

Patients should be regularly reassessed by their prescriber/health care provider, so that the dosage of RESPIDAY remains optimal. The dose should be titrated to the lowest dose at which effective control of symptoms is maintained. When control of symptoms is maintained with the lowest recommended dosage, then the next step could include a test of inhaled corticosteroid alone.

In usual practice when control of symptoms is achieved with the twice daily regimen, titration to the lowest effective dose could include RESPIDAYgiven once daily, when in the opinion of the prescriber, a long-acting bronchodilator in combination with an inhaled corticosteroid would be required to maintain control.

Increasing use of a separate rapid-acting bronchodilator indicates a worsening of the underlying condition and warrants a reassessment of the asthma therapy.

Children under 6 years: As only limited data are available, RESPIDAY is not recommended for children younger than 6 years.

RESPIDAY Discair 320/9 mcg inhalation powder should be used as RESPIDAY maintenance therapy only. Lower strengths are available for the RESPIDAY maintenance and reliever therapy regimen (RESPIDAY Discair 160/4.5 mcg inhalation powder).

COPD
Recommended doses:
Adults: 1 inhalation twice daily.

General information
Special patient groups: There are no special dosing requirements for elderly patients. There are no data available for use of RESPIDAY in patients with hepatic or renal impairment. As budesonide and formoterol are primarily eliminated via hepatic metabolism, an increased exposure can be expected in patients with severe liver diseases.

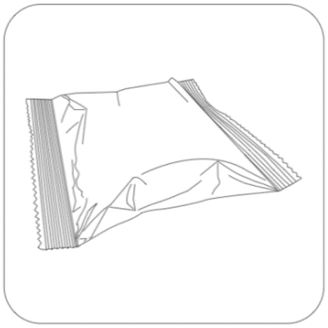
Route and method of administration:

For inhalation use.

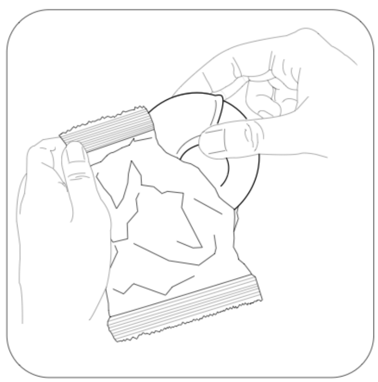
RESPIDAY is used in the form of oral inhalation. After inhalation, the mouth should be rinsed with water.
When inhaled through the inhalation device, the drug reaches the lungs. For this reason, it is important to deeply and strongly breathe through the mouthpiece of the device.
The use of the inhalation device should be indicated to the patient by the doctor or pharmacist.

Instruction for use of the inhaler

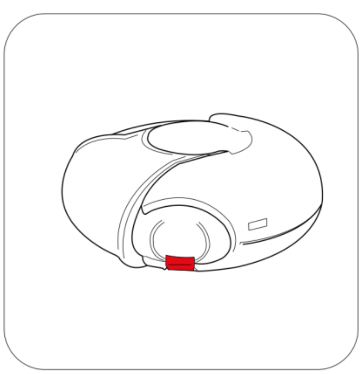
The inhalation device presented to the market in cardboard box is located in the protective packaging for safety purposes.



Before using your inhaler, remove it from the packaging as shown.



The inhalation device will be off when you remove it from the package.

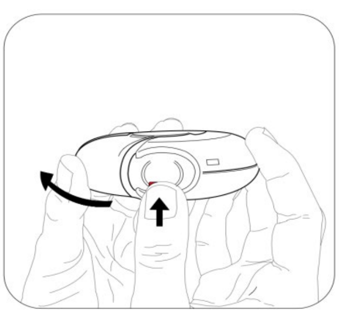
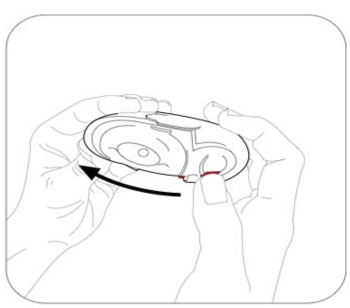


In an unused inhalation device, 60 doses of powder are separately preserved. The dose indicator shows how many doses of medication remain in the inhalation device. Each dose is precisely measured and protected according to hygienic conditions. No maintenance or refill needed.
The dose indicator at the top of the inhalation device shows how many doses remain. It's easy to use the inhalation device. The time to take the medication is shown in the following three steps.
1. Opening
2. Inhalation
3. Closing

How does the RESPIDAY inhalation device work?
Press the red button (child lock) to push the outer cover. When the outer cover is pushed, a small hole is opened in the mouthpiece and a dose of the drug is ready to be introduced. When the inhalation device is closed, the outer cover rotates to its initial position and becomes ready for the next use. The outer cover protects the inhalation device when not in use.

1. **Opening - How should you use the inhalation device?**

To open the inhalation device and make it ready for inhalation, pressing the red button will suffice to remove the outer cover. Hold the inhalation device so that the mouthpiece is facing towards you. The inhalation device is now ready for use. At each opening of the outer lid, one dose is ready for inhalation. This is seen in the dose indicator. Do not play with the outer lid to avoid losing the medication.

2. **Inhalation**

You should read this section carefully before you inhale the medicine.

• Keep the inhalation device away from your mouth. Release your breath out as much as you can. Remember - never exhale into the inhalation device.

• Rest your lips on the mouthpiece. Take a long and deep breath with your mouth. Do not breathe with your nose.

• Remove the inhaler from your mouth.

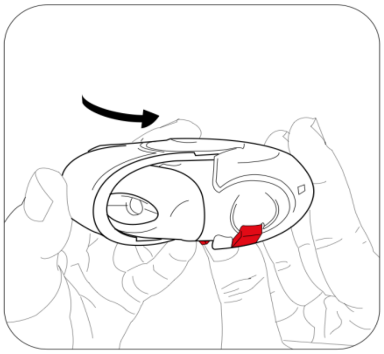
• Hold your breath for 10 seconds or as long as you can comfortably hold it.

• Breathe out slowly with your nose.

3. **Closing**

• To close the inhaler, you only need to slide the outer lid to its original position.

• The inhalation device is ready for reuse.



You may repeat the three steps above if a second inhalation is needed.

REMEMBER!

Keep the inhalation device dry and clean.

Keep the mouthpiece closed when not in use at all times.

Never exhale into the inhalation device.

Open the outer cover when you are ready to take the medication.

Do not take more than the mentioned dose.

Do not chew or bite on the mouthpiece.

Contraindications

Hypersensitivity (allergy) to the active substance budesonide, formoterol or lactose (which contains small amounts of milk protein).

Warnings and precautions

It is recommended that the maintenance dose be tapered when the long-term treatment is discontinued and the dosing should not be stopped abruptly. Complete withdrawal of inhaled corticosteroids should not be considered unless it is temporarily required to confirm the diagnosis of asthma.

If patients find the treatment ineffective, or exceed the highest prescribed dose of RESPIDAY, medical attention must be sought. Increasing use of rescue bronchodilators indicates a worsening of the underlying condition and warrants a reassessment of the asthma therapy. Sudden and progressive deterioration in control of asthma or COPD is potentially life threatening and the patient should undergo urgent medical assessment. In this situation, consideration should be given to the need for increased therapy with corticosteroids, e.g. a course of oral corticosteroids, or antibiotic treatment if an infection is present.

Patients should be advised to have their rescue inhaler available at all times.

Patients should be reminded to take their RESPIDAY maintenance dose as prescribed, even when asymptomatic.

Once asthma symptoms are controlled, consideration may be given to gradually reducing the dose of RESPIDAY. Regular review of patients as treatment is stepped down is important. The lowest effective dose of RESPIDAY should be used.

Patients should not be initiated on RESPIDAY during an exacerbation, or if they have significantly worsening or acutely deteriorating asthma.

Serious asthma-related adverse events and exacerbations may occur during treatment with RESPIDAY. Patients should be asked to continue treatment but to seek medical advice if asthma symptoms remain uncontrolled or worsen after initiation with RESPIDAY.

There are no clinical study data on RESPIDAY available in COPD patients with a pre-bronchodilator FEV1 >50% predicted normal and with a post-bronchodilator FEV1 <70% predicted normal.

As with other inhalation therapy, paradoxical bronchospasm may occur, with an immediate increase in wheezing and shortness of breath after dosing. If the patient experiences paradoxical bronchospasm, RESPIDAYshould be discontinued immediately, the patient should be assessed and an alternative therapy instituted, if necessary. Paradoxical bronchospasm responds to a rapid-acting inhaled bronchodilator and should be treated straightaway.

Systemic effects may occur with any inhaled corticosteroid, particularly at high doses prescribed for long periods. These effects are much less likely to occur with inhalation treatment than with oral corticosteroids. Possible systemic effects include Cushing's syndrome, Cushingoid features, adrenal suppression, growth retardation in children and adolescents, decrease in bone mineral density, cataract and glaucoma, and more rarely, a range of psychological or behavioural effects including psychomotor hyperactivity, sleep disorders, anxiety, depression or aggression (particularly in children).

Potential effects on bone density should be considered, particularly in patients on high doses for prolonged periods that have coexisting risk factors for osteoporosis. Long-term studies with inhaled budesonide in children at mean daily doses of 400 micrograms (metered dose) or in adults at daily doses of 800 micrograms (metered dose) have not shown any significant effects on bone mineral density. No information regarding the effect of RESPIDAY at higher doses is available.

If there is any reason to suppose that adrenal function is impaired from previous systemic steroid therapy, care should be taken when transferring patients to RESPIDAY therapy.

The benefits of inhaled budesonide therapy would normally minimise the need for oral steroids, but patients transferring from oral steroids may remain at risk of impaired adrenal reserve for a considerable time. Recovery may take a considerable amount of time after cessation of oral steroid therapy and hence oral steroid-dependent patients transferred to inhaled budesonide may remain at risk from impaired adrenal function for some considerable time. In such circumstances HPA axis function should be monitored regularly.

Prolonged treatment with high doses of inhaled corticosteroids, particularly higher than recommended doses, may also result in clinically significant adrenal suppression. Therefore additional systemic corticosteroid cover should be considered during periods of stress such as severe infections or elective surgery. Rapid reduction in the dose of steroids can induce acute adrenal crisis. Symptoms and signs which might be seen in acute adrenal crisis may be somewhat vague but may include anorexia, abdominal pain, weight loss, tiredness, headache, nausea, vomiting, decreased level of consciousness, seizures, hypotension and hypoglycaemia.

Treatment with supplementary systemic steroids or inhaled budesonide should not be stopped abruptly.

During transfer from oral therapy to RESPIDAY, a generally lower systemic steroid action will be experienced which may result in the appearance of allergic or arthritic symptoms such as rhinitis, eczema and muscle and joint pain. Specific treatment should be initiated for these conditions. A general insufficient glucocorticosteroid effect should be suspected if, in rare cases, symptoms such as tiredness, headache, nausea and vomiting should occur. In these cases a temporary increase in the dose of oral glucocorticosteroids is sometimes necessary.

To minimise the risk of oropharyngeal candida infection, the patient should be instructed to rinse their mouth out with water after inhaling the maintenance dose.

RESPIDAY should be administered with caution in patients with thyrotoxicosis, phaeochromocytoma, diabetes mellitus, untreated hypokalaemia, hypertrophic obstructive cardiomyopathy, idiopathic subvalvular aortic stenosis, severe hypertension, aneurysm or other severe cardiovascular disorders, such as ischaemic heart disease, tachyarrhythmias or severe heart failure.

Caution should be observed when treating patients with prolongation of the QTc-interval. Formoterol itself may induce prolongation of the QTc-interval.

The need for, and dose of inhaled corticosteroids should be re-evaluated in patients with active or quiescent pulmonary tuberculosis, fungal and viral infections in the airways.

Potentially serious hypokalaemia may result from high doses of β2 adrenoceptor agonists. Concomitant treatment of β2 adrenoceptor agonists with drugs which can induce hypokalaemia or potentiate a hypokalaemic effect, e.g. xanthine derivatives, steroids and diuretics, may add to a possible hypokalaemic effect of the β2 adrenoceptor agonist. Particular caution is recommended in unstable asthma with variable use of rescue bronchodilators, in acute severe asthma as the associated risk may be augmented by hypoxia and in other conditions when the likelihood for hypokalaemia adverse effects is increased. It is recommended that serum potassium levels are monitored during these circumstances.

As for all β2 adrenoceptor agonists, additional blood glucose controls should be considered in diabetic patients.

Visual disturbance may be reported with systemic and topical corticosteroid use. If a patient presents with symptoms such as blurred vision or other visual disturbances, the patient should be considered for referral to an ophthalmologist for evaluation of possible causes, which may include cataract, glaucoma or rare diseases such as central serous chorioretinopathy (CSCR), which have been reported after use of systemic and topical corticosteroids.
RESPIDAY contains lactose monohydrate (<1 mg/inhalation). This amount does not normally cause problems in lactose intolerant people. The excipient lactose contains small amounts of milk proteins, which may cause allergic reactions.

Paediatric population
It is recommended that the height of children receiving prolonged treatment with inhaled corticosteroids is regularly monitored. If growth is slowed, therapy should be re-evaluated with the aim of reducing the dose of inhaled corticosteroid to the lowest dose at which effective control of asthma is maintained. The benefits of the corticosteroid therapy and the possible risks of growth suppression must be carefully weighed. In addition, consideration should be given to referring the patient to a paediatric respiratory specialist.

Limited data from long-term studies suggest that most children and adolescents treated with inhaled budesonide will ultimately achieve their adult target height. However, an initial small but transient reduction in growth (approximately 1 cm) has been observed. This generally occurs within the first year of treatment.

Pneumonia in patients with COPD
An increase in the incidence of pneumonia, including pneumonia requiring hospitalisation, has been observed in patients with COPD receiving inhaled corticosteroids. There is some evidence of an increased risk of pneumonia with increasing steroid dose but this has not been demonstrated conclusively across all studies.

There is no conclusive clinical evidence for intra-class differences in the magnitude of the pneumonia risk among inhaled corticosteroid products.

Physicians should remain vigilant for the possible development of pneumonia in patients with COPD as the clinical features of such infections overlap with the symptoms of COPD exacerbations.

Risk factors for pneumonia in patient with COPD include current smoking, older age, low body mass index (BMI) and severe COPD.

Interaction with other Medicaments

Pharmacokinetic interactions

Potent inhibitors of CYP3A4 (e.g. ketoconazole, itraconazole, voriconazole, posaconazole, clarithromycin, telithromycin, nefazodone and HIV protease inhibitors) are likely to markedly increase plasma levels of budesonide and concomitant use should be avoided. If this is not possible the time interval between administration of the inhibitor and budesonide should be as long as possible.

The potent CYP3A4 inhibitor ketoconazole, 200 mg once daily, increased plasma levels of concomitantly orally administered budesonide (single dose 3 mg) on average six-fold. When ketoconazole was administered 12 hours after budesonide the concentration was on average increased only three-fold showing that separation of the administration times can reduce the increase in plasma levels. Limited data about this interaction for high-dose inhaled budesonide indicates that marked increases in plasma levels (on average four fold) may occur if itraconazole, 200 mg once daily, is administered concomitantly with inhaled budesonide (single dose of 1000 micrograms).

Co-treatment with CYP3A inhibitors, including cobicistat-containing products, is expected to increase the risk of systemic side-effects. The combination should be avoided unless the benefit outweighs the increased risk of systemic corticosteroid side-effects, in which case patients should be monitored for systemic corticosteroid side-effects.[1]

<u>Pharmacodynamic interactions</u>

β adrenergic blockers can weaken or inhibit the effect of formoterol. A fixed-dose combination therapy of budesonide and formoterol fumarate dihydrate should therefore not be given together with β adrenergic blockers (including eye drops) unless there are compelling reasons.

Concomitant treatment with quinidine, disopyramide, procainamide, phenothiazines, antihistamines (terfenadine) and tricyclic antidepressants can prolong the QTc-interval and increase the risk of ventricular arrhythmias.

In addition L-Dopa, L-thyroxine, oxytocin and alcohol can impair cardiac tolerance towards β_2 sympathomimetics.

Concomitant treatment with monoamine oxidase inhibitors including medicinal products with similar properties such as furazolidone and procarbazine may precipitate hypertensive reactions.

There is an elevated risk of arrhythmias in patients receiving concomitant anaesthesia with halogenated hydrocarbons.

Concomitant use of other β adrenergic medicinal products and anticholinergic medicinal products can have a potentially additive bronchodilating effect.

Hypokalaemia may increase the disposition towards arrhythmias in patients who are treated with digitalis glycosides.

Budesonide and formoterol have not been observed to interact with any other medicinal products used in the treatment of asthma.

<u>Paediatric population</u>

Interaction studies have only been performed in adults.

Pregnancy and lactation

Pregnancy

For a fixed-dose combination therapy of budesonide and formoterol fumarate dihydrate or the concomitant treatment with formoterol and budesonide, no clinical data on exposed pregnancies are available.

There are no adequate data from use of formoterol in pregnant women.

During pregnancy, a fixed-dose combination therapy of budesonide and formoterol fumarate dihydrate should only be used when the benefits outweigh the potential risks. The lowest effective dose of budesonide needed to maintain adequate asthma control should be used.

Breast-feeding

Budesonide is excreted in breast milk. However, at therapeutic doses no effects on the suckling child are anticipated. It is not known whether formoterol passes into human breast milk. In rats, small amounts of formoterol have been detected in maternal milk. Administration of a fixed-dose combination therapy of budesonide and formoterol fumarate dihydrate to women who are breast-feeding should only be considered if the expected benefit to the mother is greater than any possible risk to the child.

Fertility

There is no data available on the potential effect of budesonide on fertility.

Side effects

Adverse reactions, which have been associated with budesonide or formoterol, are given below and listed by system organ class:

System Organ Class	**Frequency**	**Adverse reaction**
Infections and infestations	Common	Candida infections in the oropharynx, pneumonia (in COPD patients)
Immune system disorders	Rare	Immediate and delayed hypersensitivity reactions, e.g. exanthema, urticaria, pruritus, dermatitis, angioedema and anaphylactic reaction
Endocrine disorders	Very rare	Cushing´s syndrome, adrenal suppression, growth retardation, decrease in bone mineral density
Metabolism and nutrition disorders	Rare	Hypokalaemia
	Very rare	Hyperglycaemia

Psychiatric disorders	Uncommon	Aggression, psychomotor hyperactivity, anxiety, sleep disorders
	Very rare	Depression, behavioural changes (predominantly in children)
Nervous system disorders	Common	Headache, tremor
	Uncommon	Dizziness
	Very rare	Taste disturbances
Eye disorders	Very rare	Cataract and glaucoma
	Uncommon	Vision, blurred
Cardiac disorders	Common	Palpitations
	Uncommon	Tachycardia
	Rare	Cardiac arrhythmias, e.g. atrial fibrillation, supraventricular tachycardia, extrasystoles
	Very rare	Angina pectoris. Prolongation of QTc-interval
Vascular disorders	Very rare	Variations in blood pressure
Respiratory, thoracic and mediastinal disorders	Common	Mild irritation in the throat, coughing, hoarseness
	Rare	Bronchospasm
	Very rare	Paradoxical bronchospasm
Gastrointestinal disorders	Uncommon	Nausea
Skin and subcutaneous tissue disorders	Uncommon	Bruises
Musculoskeletal and connective tissue disorders	Uncommon	Muscle cramps

Description of selected adverse reactions

Candida infection in the oropharynx is due to active substance deposition. Advising the patient to rinse the mouth out with water after each dose will minimise the risk. Oropharyngeal Candida infection usually responds to topical anti-fungal treatment without the need to discontinue the inhaled corticosteroid.

Paradoxical bronchospasm may occur very rarely, affecting less than 1 in 10,000 people, with an immediate increase in wheezing and shortness of breath after dosing. Paradoxical bronchopasm responds to a rapid-acting inhaled bronchodilator and should be treated straightaway. RESPIDAY should be discontinued immediately, the patient should be assessed and an alternative therapy is instituted if necessary.

Systemic effects of inhaled corticosteroids may occur, particularly at high doses prescribed for long periods. These effects are much less likely to occur than with oral corticosteroids. Possible systemic effects include Cushing´s syndrome, Cushingoid features, adrenal suppression, growth retardation in children and adolescents, decrease in bone mineral density, cataract and glaucoma. Increased susceptibility to infections and impairment of the ability to adapt to stress may also occur. Effects are probably dependent on dose, exposure time, concomitant and previous steroid exposure and individual sensitivity.

Treatment with β_2 adrenoceptor agonists may result in an increase in blood levels of insulin, free fatty acids, glycerol and ketone bodies.

Symptoms and treatment of overdose

An overdose of formoterol would likely lead to effects that are typical for β_2 adrenoceptor agonists: tremor, headache, palpitations. Symptoms reported from isolated cases are tachycardia, hyperglycaemia, hypokalaemia, prolonged QTc-interval, arrhythmia, nausea and vomiting. Supportive and symptomatic treatment may be indicated. A dose of 90 micrograms administered during three hours in patients with acute bronchial obstruction raised no safety concerns.

Acute overdose with budesonide, even in excessive doses, is not expected to be a clinical problem. When used chronically in excessive doses, systemic glucocorticosteroid effects, such as hypercorticism and adrenal suppression, may appear.

If RESPIDAY therapy has to be withdrawn due to overdose of the formoterol component of the medicinal product, provision of appropriate inhaled corticosteroid therapy must be considered.

Effects on ability to drive and use machines

RESPIDAY has no or negligible influence on the ability to drive and use machines.

Storage condition

Store at room temperature below 30°C. Keep the container tightly closed, in order to protect from moisture.

Dosage forms and packaging available
RESPIDAY Discair 160/4.5 mcg inhalation powder

RESPIDAY is contained in the inhalation device in Alu / Alu blister packages containing 60 doses and 2 devices x 60 doses of inhalation powder.

RESPIDAY Discair 320/9 mcg inhalation powder

RESPIDAY is contained in the inhalation device in Alu / Alu blister packages containing 60 doses of inhalation powder.

Manufacturer
Neutec Inhaler Ilac Sanayi ve Ticaret A.S.,

Turkey, Sakarya, Arifiye,

1. Organize Sanayi Bolgesi, 2 Nolu Yol, No:3

TURKEY

Date of revision

May 2024

Times New Roman: Size 7